

Chapter 6

AI Applications

Overview

6.1 Expert Systems: Characteristics, Architecture, Development and Various Applications

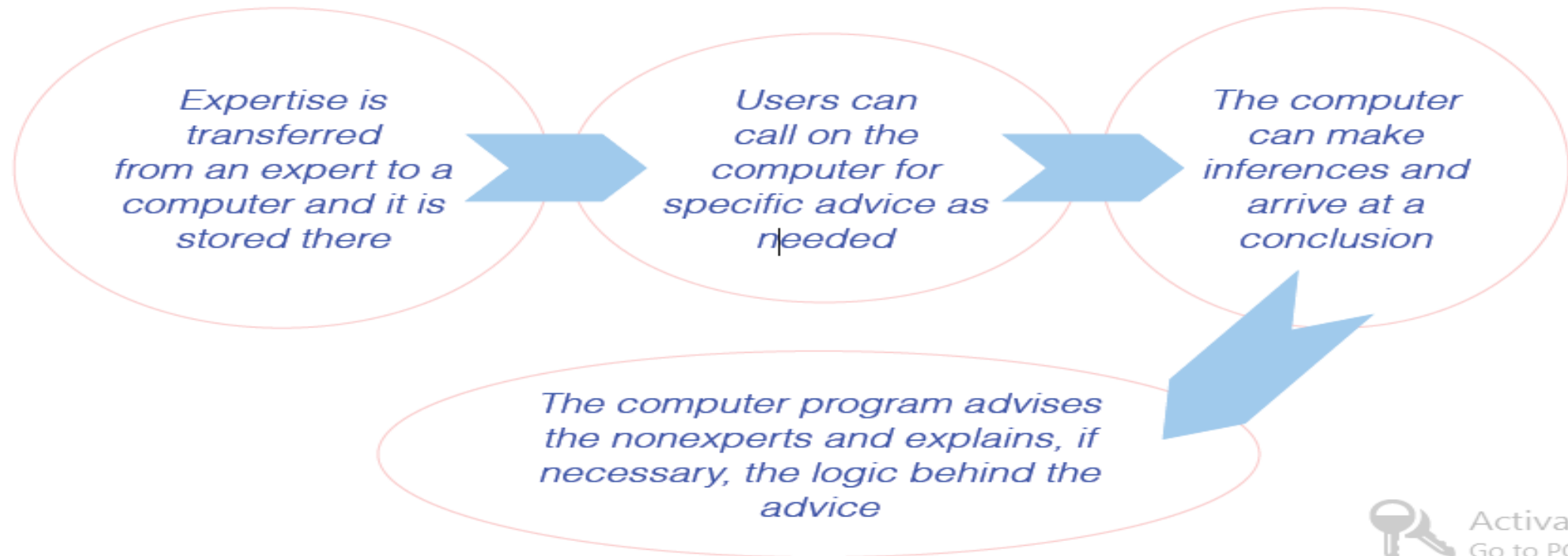
6.2 NLP: Level of Analysis, Challenges, Modern Approaches and Applications

6.3 Robotics & Computer Vision: Fundamental, Components and Applications

6.4 Sustainable AI Development

6.1 Expert Systems: Characteristics, Architecture, Development and Various Applications

- The expert systems are the computer applications developed to solve complex problems in a particular domain, at the level of extraordinary human intelligence and expertise.



Characteristics of Expert Systems

- High performance
- Domain-Specific Knowledge
- Reasoning Capability
- Reliable
- Highly responsive
- User Friendly Interface

When to Use an Expert System

- Provide a high potential payoff or significantly reduced downside risk.
- Capture and preserve irreplaceable human expertise.
- Provide expertise needed at a number of locations at the same time or in a hostile environment that is dangerous to human health.
- Provide expertise that is expensive or rare.
- Develop a solution faster than human experts can.
- Provide expertise needed for training and development to share the wisdom of human experts with a large number of people.

Human Expert Vs Expert System

Factor	Human Expert	Expert System
Time (can be obtained)	Working days only	Anytime
Geography	Local	Anywhere
Safety	Cannot be replaced	Can be replaced
Damages	Yes	No
Speed and Efficiency	Changes	Consistent
Cost	High	Intermediate

Examples of Expert System

- MYCIN would attempt to diagnose patients based on reported symptoms and medical test results.
- The program could request further information concerning the patient, as well as suggest additional laboratory tests, to arrive at a probable diagnosis, after which it would recommend a course of treatment.
- If requested, MYCIN would explain the reasoning that led to its diagnosis and recommendation.

Examples of Expert System

- CALEX is a blackboard based integrated system for agricultural management, developed at University of California. CALEX can be used by growers, pest control advisors, consultants and other managers.
- DENDRAL's primary aim was to help organic chemists in identifying unknown organic molecules, by analyzing their mass spectra and using knowledge of chemistry.
- PXDES is an expert system that is used to determine the type and level of lung cancer. To determine the disease, it takes a picture from the upper body, which looks like the shadow. This shadow identifies the type and degree of harm.
- CaDeT is a diagnostic support system that can detect cancer at early stages.
- XCON is a manufacturing expert system used to configure computer systems based on customer requirements.

Components/Architecture of Expert Systems

- The components of Expert System include :

- Knowledge Base

- Inference Engine

- User Interface

Components/Architecture of Expert Systems

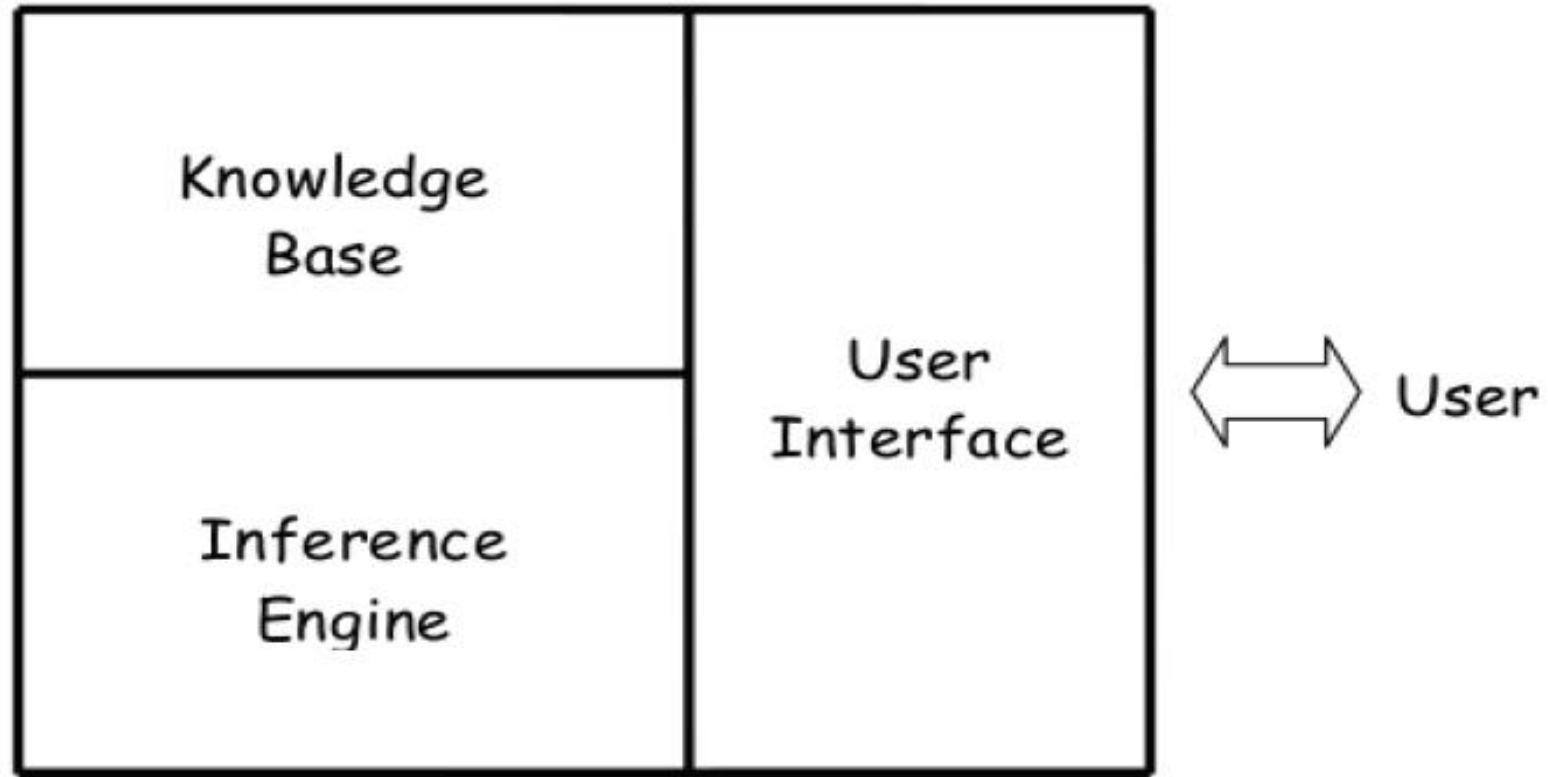


Fig: Block Diagram of expert system

Knowledge Base

- Knowledge is required to exhibit intelligence.
- The success of any ES depends upon the collection of highly accurate and precise knowledge.

Components of Knowledge Base

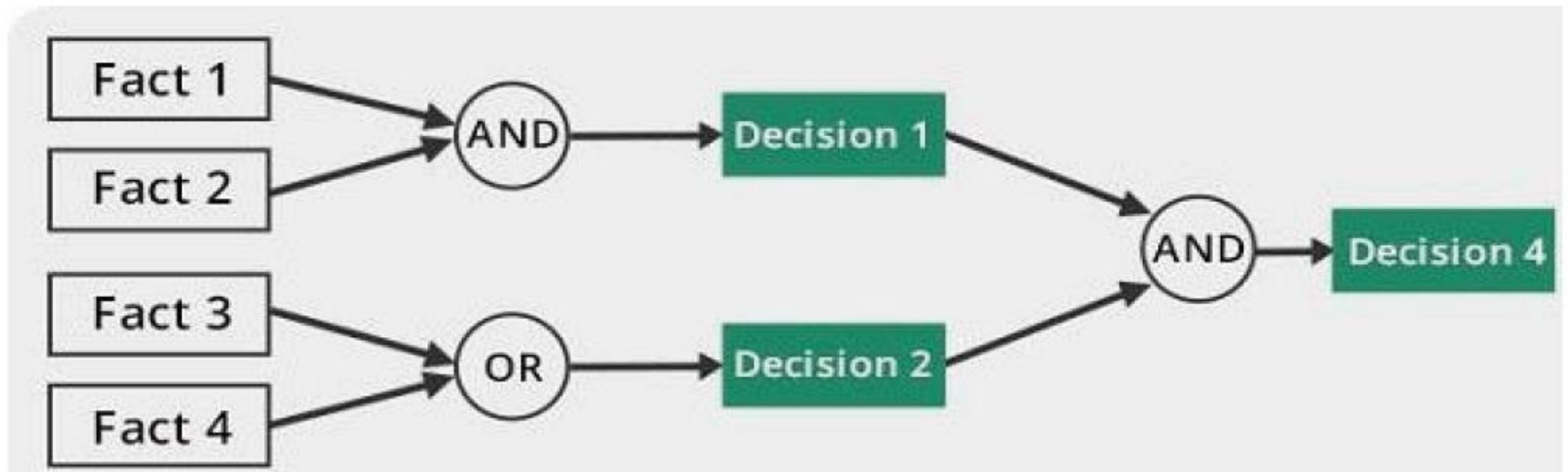
- The knowledge base of an ES is a store of both, factual and heuristic knowledge.
 - Factual Knowledge – It is the information widely accepted by the Knowledge Engineers and scholars in the task domain.
 - Heuristic Knowledge – It is about practice, accurate judgment, one's ability of evaluation, and guessing.

Inference Engine

- In case of knowledge-based Expert System, the Inference Engine acquires and manipulates the knowledge from the knowledge base to arrive at a particular solution.
- In case of rule based Expert System, it –
 - Applies rules repeatedly to the facts, which are obtained from earlier rule application.
 - Adds new knowledge into the knowledge base if required.
 - To recommend a solution, the Inference Engine uses the following strategies:
 - Forward Chaining
 - Backward Chaining

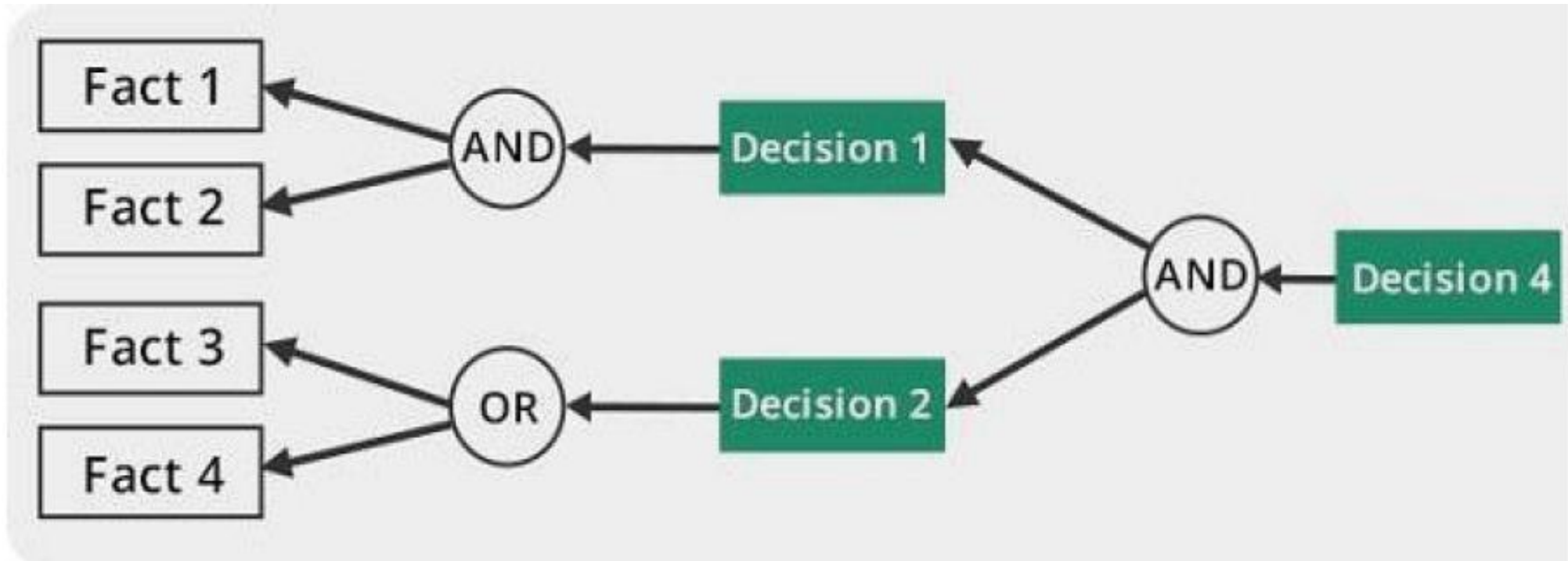
Forward Chaining

- It is a strategy of an expert system to answer the question, “What can happen next?”



Backward Chaining

- With this strategy, an expert system finds out the answer to the question, “Why this happened?”.



User Interface

- User interface provides interaction between user of the ES and the ES itself.
- It is generally Natural Language Processing so as to be used by the user who is well-versed in the task domain.
- The user of the ES need not be necessarily an expert in Artificial Intelligence.

Stages of Expert System Development

Phases in the Development of an Expert System

1. Problem Definition

- Identify the problem domain.
- Determine whether an expert system is suitable for the problem.

2. Knowledge Acquisition

- Collect domain-specific knowledge from human experts, documents, or databases.
- Involves interviews, observations, or using automated tools.

3. Knowledge Representation

- Organize the acquired knowledge into a format the system can use.
- Common representations:
 - **Rules** (IF-THEN)
 - **Semantic networks**
 - **Frames**

Stages of Expert System Development

4. Design of Inference Engine

- Develop the reasoning mechanism that applies logical rules to the knowledge base to infer conclusions.
- Two main reasoning approaches:
 - Forward chaining (data-driven)
 - Backward chaining (goal-driven)

5. Implementation

- Build the system using appropriate tools (e.g., CLIPS, Prolog, Python).
- Develop the User Interface and integrate the Knowledge Base and Inference Engine.

6. Testing and Validation

- Evaluate the system's performance by comparing its outputs to expert decisions.
- Adjust rules or knowledge as needed.

7. Maintenance

- Update the system as new knowledge becomes available or when domain rules change.
- Ensure the system remains relevant and accurate.

Expert Systems Limitations

- Limitations of the technology
- Difficult knowledge acquisition
- Expert Systems are difficult to maintain
- High development costs

Benefits of Expert Systems

- Availability – They are easily available due to mass production of software.
- Speed – They offer great speed. They reduce the amount of work an individual puts in.
- Less Error Rate – Error rate is low as compared to human errors.
- Reducing Risk – They can work in the environment dangerous to humans.
- Steady response – They work steadily without getting emotional, tensed or fatigued.

Applications of Expert System

Application	Description
Design Domain	Camera lens design, automobile design.
Medical Domain	Diagnosis Systems to deduce cause of disease from observed data, conduction medical operations on humans.
Monitoring Systems	Comparing data continuously with observed system or with prescribed behavior such as leakage monitoring in long petroleum pipeline.
Process Control Systems	Controlling a physical process based on monitoring.
Knowledge Domain	Finding out faults in vehicles, computers.
Finance/Commerce	Detection of possible fraud, suspicious transactions, stock market trading, Airline scheduling, cargo scheduling.



6.2 NLP: Level of Analysis, Challenges, Modern Approaches and Applications

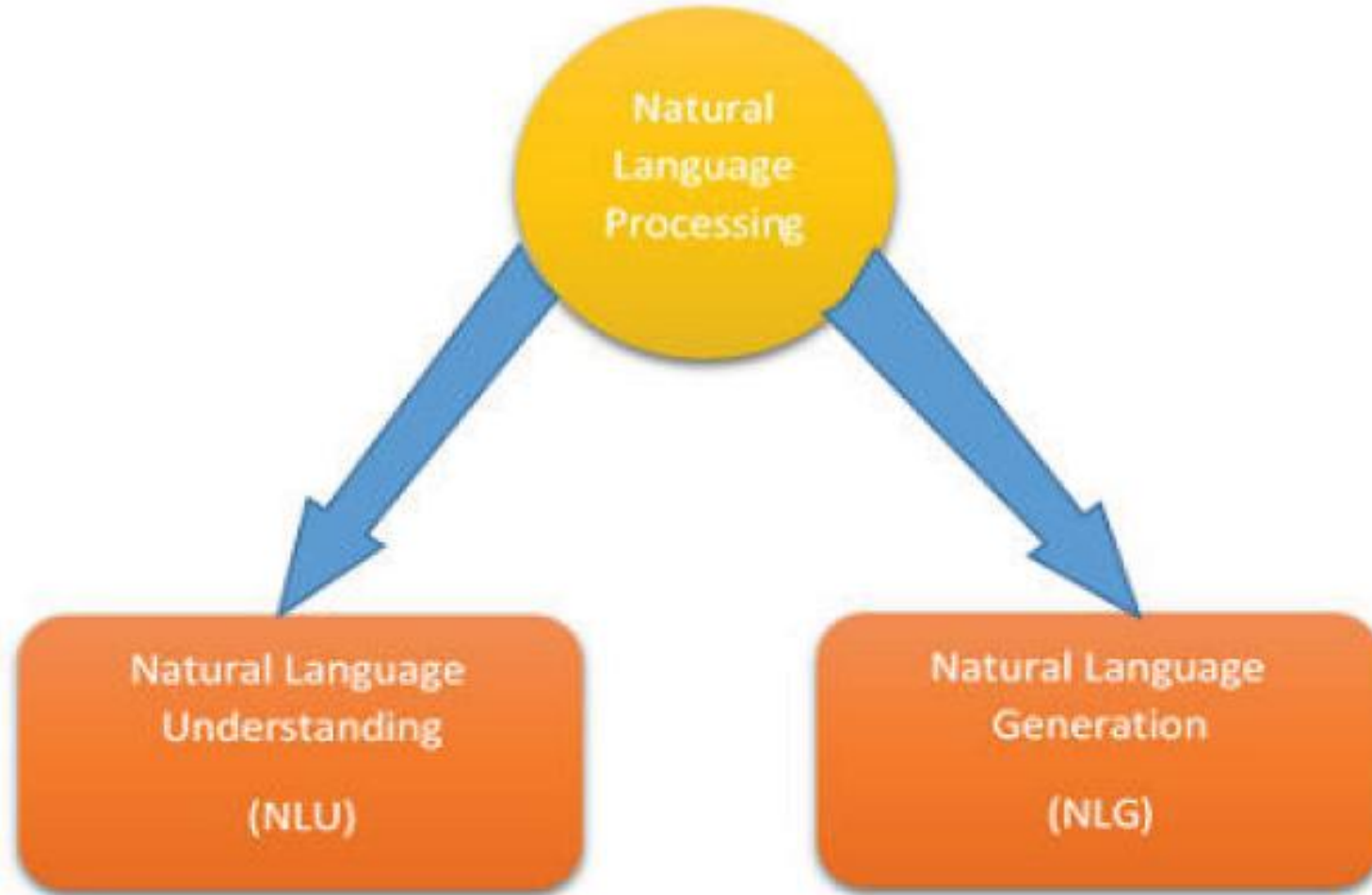
Natural Language

- In philosophy of language, a natural language or ordinary language is any language that has evolved naturally in humans through use and repetition without conscious planning or premeditation.
- Natural languages can take different forms, such as speech or signing.
- They are distinguished from constructed and formal languages such as those used to program computers or to study logic.
- In computing, natural language refers to a human language such as English, Russian, German, or Japanese as distinct from the typically artificial command or programming language with which one usually talks to a computer.
- Humans perceive and communicate through their five basic senses of sight, hearing, touch, smell and taste, and their ability to generate meaningful utterances.
- Developing programs that understand a natural language is a difficult problem.

Natural Language

- Developing programs to understand natural language is important in AI because a natural form of communication with systems is essential.
- AI programs must be able to communicate with their human counterparts in a natural way, and natural language is one of the most important mediums for that purpose.
- So, Natural Language Processing (NLP) is the field that deals with the computer processing of natural languages, mainly evolved by people working in the field of Artificial Intelligence.

Components in NLP



Natural Language Understanding

- It deals with machine reading comprehension : ability to read text, to do process on it, and understand its meaning.
- It is involved with mapping the given inputs in useful representations and analyzing the different aspects of the language.
- Many words have several meaning such as can, bear, fly, etc and sentences have different meanings in different contexts.
- Understanding the language is not only the transmission of words. It also requires inference about the speakers' goal, knowledge as well as the context of the interaction.

Difficulties in NLU

- Natural Language has an extremely rich form and structure.
- It is very ambiguous.
- Different levels of ambiguity :
 - Lexical ambiguity
 - Syntax Level (Syntactic) ambiguity
 - Referential ambiguity

Ambiguity in Natural Language

1. Lexical Ambiguity (Word-Level Ambiguity)

- When a single word has multiple meanings. Also called word-sense ambiguity.

Example 1: "She went to the bank."

- The word bank can mean: Financial institution, River side

Example 2: "He saw a bat."

- The word Bat can mean: Animal, Sports equipment

2. Syntactic Ambiguity (Structural Ambiguity)

- When a sentence has multiple grammatical structures, leading to different meanings.

Example 1: "I saw the man with the telescope." has two interpretations:

- I used a telescope to see the man.
- The man had a telescope.

Example 2:

- "Visiting relatives can be annoying." has two interpretations
- When you visit relatives
- Relatives who visit you

Ambiguity in Natural Language

3. Referential Ambiguity

- When it is unclear which noun, a pronoun refers to.
- Also called coreference ambiguity.

Example 1:

- "Geeta told Anu that she won the award."
- Who won the award?
- Geeta? or Anu?

Example 2:

- "The laptop doesn't fit in the bag because it is too small."
- What is too small?
- Laptop? or Bag?

Natural Language Generation

- NLG also referred to as text generation, is a subfield of natural language processing.
- Natural Language Generation (NLG) is the natural language processing task of generating natural language from a machine representation system such as a knowledge base or a logical form.
- NLG system is like a translator that converts a computer-based representation into a natural language representation.

NLG includes

- Text planning : - Retrieving the relevant content from database. Here database can include vocabulary, sentences, knowledge, sample data and many more.
- Sentence planning : - We get our content using text planning now next step to do is choosing required words and forming meaningful sentence setting the words in right grammatical way.
- Text realization :- We have all the thing need to create actual text in human language.

Levels of knowledge used in Language Understanding

- The component forms of knowledge needed for an understanding of natural languages are sometimes classified according to the following levels.

Phonological:

- Relates sound to the words we recognize. A phoneme is the smallest unit of the sound.

Morphological:

- This is lexical (Linguistic unit) knowledge which relates to the word construction from basic units called morphemes. A morpheme is the smallest unit of meaning.
- Eg:- friend + ly = friendly.

Levels of knowledge used in Language Understanding

Syntactic:

- This knowledge relates to how words are put together or structured together to form grammatically correct sentences in the language.

Semantic :

- This knowledge is concerned with the meanings of words and phrases and how they combine to form sentence meaning.

Pragmatic:

- This is high – level knowledge which relates to the use of sentences in different contexts and how the context affects the meaning of the Sentence.

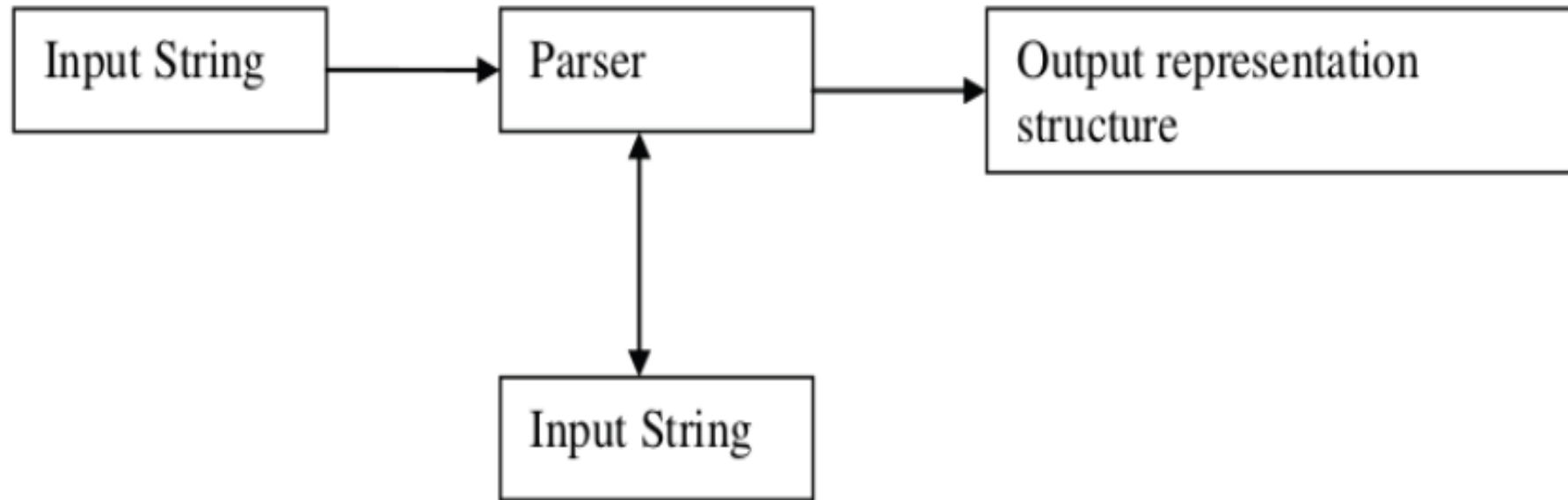
World :

- Includes the knowledge of the physical world, the world of human social interaction, and the roles of goals and intentions in communication.

Basic Parsing Techniques

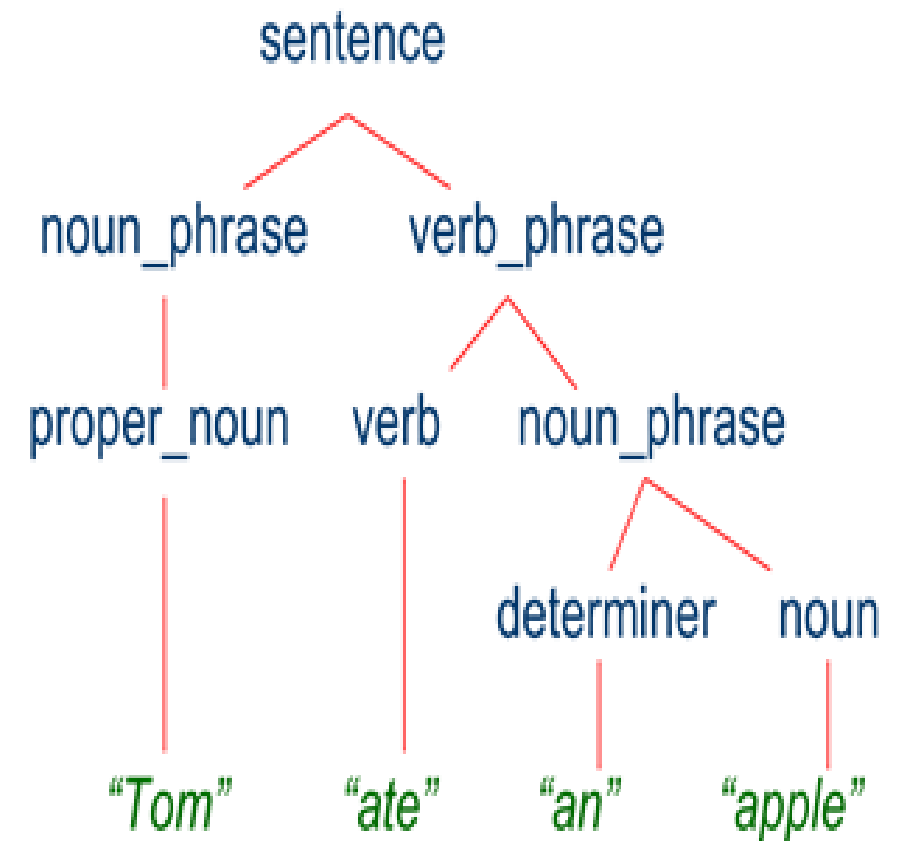
- Before the meaning of a sentence can be determined, the meanings of its constituent parts must be established.
- This requires knowledge of the structure of the sentence, the meaning of the individual words and how the words modify each other.
- The process of determining the syntactical structure of a sentence is known as parsing.
- Parsing is the process of analyzing a sentence by taking it apart word – by – word and determining its structure from its constituent parts and sub parts.
- The structure of a sentence can be represented with a syntactic tree.
- When given an input string, the lexical parts or terms (root words), must first be identified by type and then the role they play in a sentence must be determined.
- These parts can be combined successively into larger units until a complete tree has been computed.

Basic Parsing Techniques



Example sentence “Tom ate an apple”

sentence -> noun_phrase, verb_phrase
noun_phrase -> proper_noun
noun_phrase -> determiner, noun
verb_phrase -> verb, noun_phrase
proper_noun -> [Tom]
noun -> [apple]
verb -> [ate]
determiner -> [an]



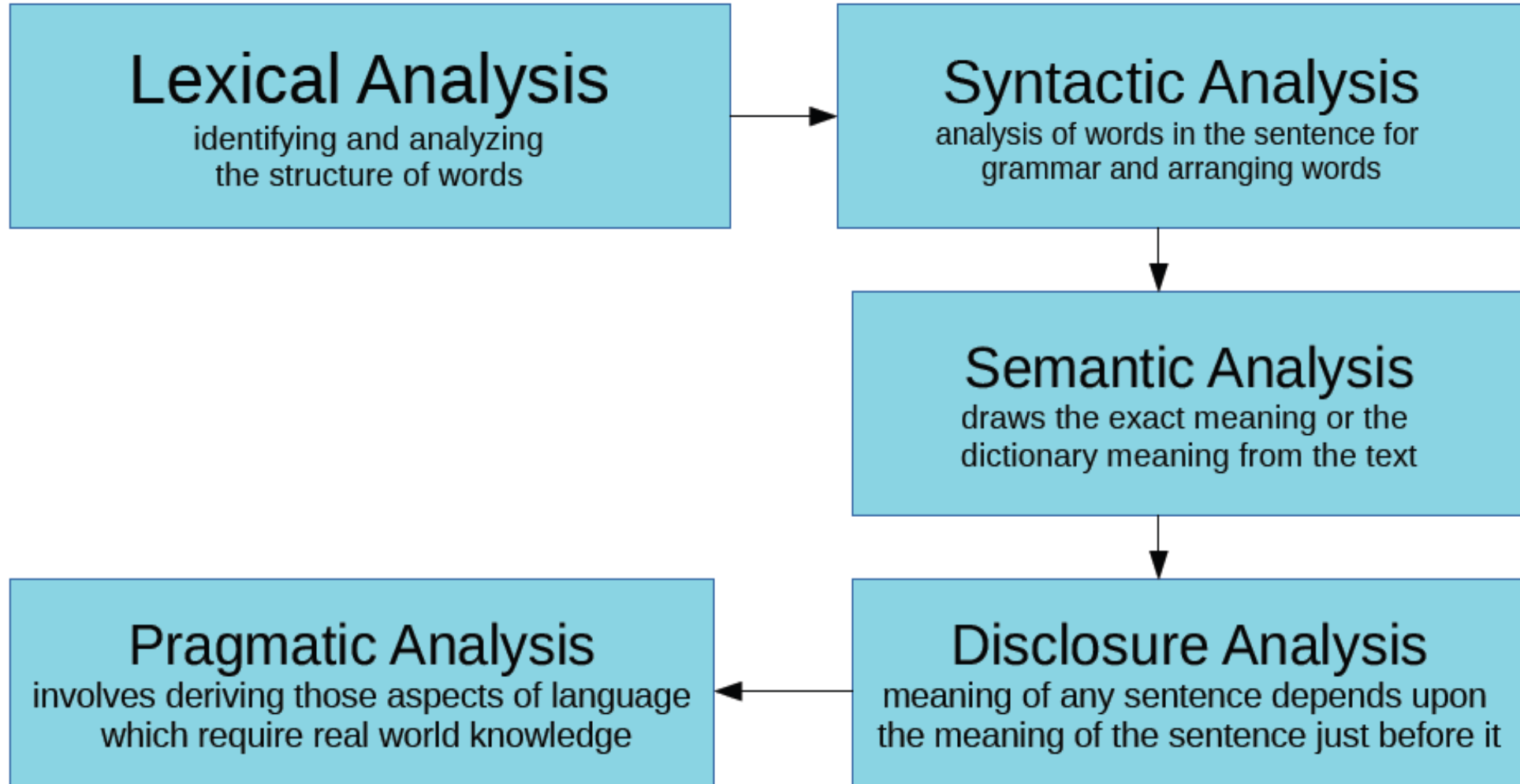
A. Top-Down parsing

- Using Top-Down technique, parser searches for a parse tree by trying to build from the root node S down to the leaves.
- The algorithm starts by assuming the input can be derived by the designated start symbol S .
- The next step is to find the tops of all the trees which can start with S , by looking on the grammar rules with S on left hand side, all the possible trees are generated.
- Top-Down parsing is goal directed search technique.

B. Bottom – Up

- Bottom – Up parsing starts with the words of input and tries to build trees from the words up, again by applying rules from the grammar one at a time.
- The parse is successful if the parse succeeds in building a tree rooted in the start symbol S that covers all of the input.
- Bottom-up parsing is a data directed search.

Steps used in NLP



Steps used in NLP

1. Lexical and Morphological Analysis

- It focuses on identifying and processing words (or lexemes) in a text.
- It breaks down the input text into individual tokens that are meaningful units of language such as words or phrases.

Key tasks in Lexical analysis:

- Tokenization: Process of dividing a text into smaller chunks called tokens. For example: the sentence "I love programming" would be tokenized into ["I", "love", "programming"].
- Part-of-Speech Tagging: Assigning parts of speech such as noun, verb, adjective to each token in the sentence. This helps us to understand grammatical roles of words in the context.
- Example: Consider the sentence: "I am reading a book."
- Tokenization: Sentence is broken down into individual tokens or words: ["I", "am", "reading", "a", "book"]
- Part-of-Speech Tagging: Each token is assigned a part of speech: ["I" → Pronoun, "am" → Verb, "reading" → Verb, "a" → Article, "book" → Noun]

Steps used in NLP

- Morphological Analysis
- It deals with morphemes which are the smallest units of meaning in a word.
- It is important for understanding structure of words and their parts by identifying free morphemes (independent words like "cat") and bound morphemes (like prefixes or suffixes e.g. "un-" or "-ing").

Key tasks in morphological analysis:

- Stemming: Reducing words to their root form like "running" to "run".
- Lemmatization: Converting words to their base or dictionary form considering the context like "better" becomes "good".

Steps used in NLP

2. Syntactic Analysis (Parsing)

- Syntactic Analysis helps in understanding how words in a sentence are arranged according to grammar rules.
- It ensures that the sentence follows correct grammar which makes the meaning clearer by constructing a parse tree.

Key components of syntactic analysis include:

- Part Of Speech (POS) Tagging: Assigning parts of speech (noun, verb, adjective) to words in a sentence.
- Ambiguity Resolution: Handling words that have multiple meanings (e.g "book" can be a noun or a verb).
- Examples:
- Consider the following sentences:
 - Correct Syntax: "John eats an apple."
 - Incorrect Syntax: "Apple eats John an."

Steps used in NLP

3. Semantic Analysis

- Semantic Analysis focuses on understanding meaning behind words and sentences.
- It ensures that the text is not only grammatically correct but also logically coherent and contextually relevant.
- It aims to understand dictionary definitions of words and their usage in context and also find whether the arrangement of words in a sentence makes logical sense.

Key Tasks in Semantic Analysis:

- Named Entity Recognition (NER): It identifies and classifies entities such as names of people, locations, organizations, dates and more. These entities provide important meaning in the text and help in understanding the context. For example: in the sentence "Tesla announced its new electric vehicle in California," NER would identify "Tesla" as an organization and "California" as a location.
- Word Sense Disambiguation (WSD): Many words have multiple meanings depending on the context in which they are used. It identifies the correct meaning of a word based on its surrounding text. For example: "bank" can refer to a financial institution or the side of a river.

Example of Semantic Analysis:

- "Apple eats a John." while grammatically correct this sentence doesn't make sense semantically because an apple cannot "eat" a person.

Steps used in NLP

4. Discourse Integration

- It is the process of understanding how individual sentences or segments of text connect and relate to each other within a broader context.
- This phase ensures that the meaning of a text is consistent and coherent across multiple sentences or paragraphs.
- It is important for understanding long or complex texts where meaning focuses on previous statements.

Key aspects of discourse integration:

- **Anaphora Resolution:** Anaphora refers to the use of pronouns or other references that depend on earlier parts of the text. For example: in the sentence "Taylor went to the store. She bought groceries" pronoun "She" refers back to "Taylor".
- **Contextual References:** Many words or phrases can only be fully understood when considered in the context of following sentences. For example: "It was a great day" is clearer when you know what event or situation is being discussed.

Example of Discourse Integration:

- "Taylor went to the store to buy some groceries. She realized she forgot her wallet." Understanding that "Taylor" is the antecedent of "she" is important for understanding sentence's meaning.
- "This is unfair!" helps in understand what "this" refers to we need to identify following sentences. Without context statement's meaning remains unclear.

Steps used in NLP

5. Pragmatic Analysis

- Pragmatic analysis helps in understanding the deeper meaning behind words and sentences by looking beyond their literal meanings.
- While semantic analysis looks at the direct meaning it considers the speaker's or writer's intentions, tone and context of the communication.

Key tasks in pragmatic analysis:

- Understanding Intentions: Sometimes language doesn't mean what it says literally. For example: when someone asks "Can you pass the salt?" it's not about ability but a polite request. It helps to understand true intention behind such expressions.
- Figurative Meaning: Language often uses idioms or metaphors that can't be taken literally.

Examples of Pragmatic Analysis:

- "Hello! What time is it?" here it might be a straightforward request for the current time but it could also imply concern about being late.

Challenges of NLP

- Natural Language contain infinity of different sentences.
- No matter how many sentences a person has heard or seen, new ones can always be produced.
- Also, there is much ambiguity in a natural language.
- Many words have several meanings and sentences can have different meanings in different contexts.
- This makes the creation of programs that understand a natural language, one of the most challenging tasks in AI.

Applications of NLP:

- Voice recognition software: Translates speech into input for word processors or other applications;
- Text-to-speech synthesizers: Read text aloud for users such as the visually-impaired;
- Grammar and style checkers: Analyze text in an attempt to highlight errors of grammar or usage;
- Machine translation: Systems which automatically render a document such as a web page in another language.

Embeddings & Modern NLP Models

❖ Representing text as numerical data is a fundamental requirement for any AI or machine learning model, as these models can only process numbers, not raw text. This process is called **Text Vectorization** or **Word Embedding**.

A. One-hot Encoding

B. Bag-of-Words (BoW)

C. TF-IDF (Term Frequency-Inverse Document Frequency)

D. Word2vec

One-Hot Encoding

- ❖ One-hot encoding represents the simplest approach to converting text into numerical form, treating each unique word in a vocabulary as an independent, atomic symbol. A word is represented by a vector where the dimension corresponding to its index is 1, and all other dimensions are 0. The approach is only useful for small, controlled vocabularies and categorical feature encoding.

- ❖ **Example:** Vocabulary: { "The", "cat", "sits", "on", "mat" }

- "cat" $\rightarrow [0, 1, 0, 0, 0]$
- "sits" $\rightarrow [0, 0, 1, 0, 0]$ and so on for other words

	I	love	NLP	is	future
I love NLP	1	1	1	0	0
NLP is future	0	0	1	1	1

- ❖ **Advantages:**

- Simple and easy to implement.
- Maintains clear distinction between words.

- ❖ **Limitations:**

- High dimensionality for large vocabularies.
- Most values are zero, making it inefficient for storage and computation.
- No semantic relationship captured — all words are equally distant.

Bag-of-Words(BoW)

- ❖ Bag-of-Words (BoW) is a very simple algorithm that ignores the position of words in the text and only considers this frequency property. It converts text into a numerical feature vector that machine learning models can process.
- ❖ The BoW model:
 - Treats a text (sentence, paragraph, or document) as a bag (collection) of words.
 - Ignores grammar and word order, but keeps word frequency.
 - Represents text as a vector based on how many times each word appears.

Example: Suppose we have two sentences:

1. "The cat sat on the mat"
2. "The dog sat on the log"



Word	Index
the	0
cat	1
sat	2
on	3
mat	4
dog	5
log	6

Step 1: Build vocabulary



Sentence	Vector
1	[2, 1, 1, 1, 1, 0, 0]
2	[2, 0, 1, 1, 0, 1, 1]

Step 2: Vector representation (BoW)

Each value shows the frequency of a word in the sentence.

Word Embedding Models

- ❖ Word embedding is the feature learning technique where words from the vocabulary are mapped to vectors of real numbers capturing the contextual hierarchy.
- ❖ Word embeddings are prediction based, and they use shallow neural networks to train the model that will lead to learning the weight and using them as a vector representation.
- ❖ Some word embedding approaches
 - **Word2Vec** (Skip-Gram, Continuous Bag of Words (CBOW)) - (2013 - by Tomas Mikolov et al. at Google)
 - **FastText** (character n-grams) - (2016 (by Piotr Bojanowski, Edouard Grave, Armand Joulin, and Tomas Mikolov at Facebook AI Research)
 - **Glove** (matrix factorization technique on word context matrix) - (2014 by Jeffrey Pennington, Richard Socher, and Christopher Manning at Stanford)

Word2Vec

- Word2Vec is a powerful word embedding technique in NLP that represents words as dense numerical vectors in a continuous vector space. Unlike Bag-of-Words or TF-IDF (which are based on *frequency counts*), Word2Vec captures semantic relationships i.e. words with similar contexts have similar vector representations.
- Example:** “King” and “Queen” appear in similar contexts (“royal,” “palace,” “crown”), so Word2Vec places them close in vector space.

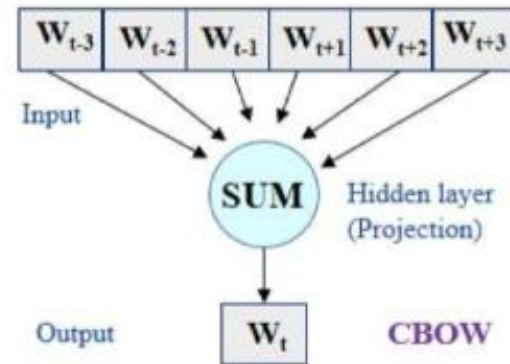
- Two Main Architectures**

- A. CBOW (Continuous Bag of Words)**

- Predicts a word from its context (surrounding words).
- Example:** Given “the ___ sat on the mat,” predict “cat.”

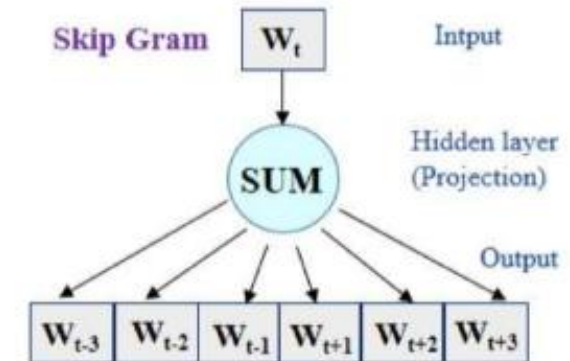
- B. Skip-Gram**

- Predicts context words from a target word.
- Example:** Given “cat,” predict words like “the,” “sat,” “on,” “mat.”



Predicts a target word
given its context words

- Input:** Multiple context words (averaged)
- Output:** Probability distribution for target word



Predicts context words
given a target word

- Input:** Single target word
- Output:** Probability distribution for context words

Modern NLP Models

Model	Typical Applications
BERT (Bidirectional Encoder Representations from Transformers)	Question-Answer, Sentiment Analysis
RoBERTa	Text Classification, Question-Answer
GPT(2/3/4)	Chatbots, Summarization
T5(Text-to-Text Transfer Transformer)	Translation, Summarization
XLNet	Question-Answer, Text Classification
ALBERT	Mobile & Low resource NLP
ELECTRA	Classification, NER
DistilBERT	Real-time NLP tasks
Sentence-BERT (SBERT)	Text Similarity, Clustering
mBERT	Low-resources languages

6.3. Robotics & Computer Vision: Fundamental, Components and Applications

- Robotics

- Robotics is a branch of Artificial Intelligence and Engineering concerned with designing, building, programming, and operating intelligent machines capable of performing tasks automatically.
- A robot is a programmable machine capable of sensing its environment, processing information, making decisions, and performing actions.
- Traditional robots used to follow predefined instructions, could not adapt to new situations & could not learn from experience.
- Whereas modern day AI-powered robots learn from data, recognizes objects, navigate unfamiliar environments, make intelligent decisions & improve performance over time.
- Example: Traditional Robot: Move from Point A to Point B using a fixed path while AI Robot is capable of finding the shortest path while avoiding unexpected obstacles.

Fundamental Components of a Robot

1. Sensors

- Sensors collect information from the environment.
- Examples: Camera, Ultrasonic Sensor, Infrared Sensor, GPS, LiDAR, Temperature Sensor

2. Controller (Brain)

- The controller processes sensor information.
- Examples: Raspberry Pi, Arduino, NVIDIA Jetson, Industrial PLC.
- Its functions are data processing, decision making, path planning & motion control.

3. Actuators

- Actuators convert electrical signals into physical movement.
- Examples: DC Motor, Servo Motor, Stepper Motor, Hydraulic Cylinder

Fundamental Components of a Robot

4. Power Supply

- Provides electrical energy.
- Examples: Batteries, Solar Cells, AC Power, Fuel Cells

5. End Effectors

- The part that performs the task.
- Examples: Robot Gripper, Welding Torch, Vacuum Suction Cup, Surgical Tool

Robot Classification

- We categorize robots primarily by mobility and workspace:

A. By Kinematic Structure (Mobility)

- Fixed/Gantry Robots: These are bolted to the floor. Think of the massive arms in automotive assembly lines. They have high precision but a limited, fixed workspace.
- Wheeled/Legged Robots: These are mobile platforms. Wheeled robots (like delivery drones) are energy-efficient on flat surfaces. Legged robots (like Boston Dynamics' Spot) are highly adaptable to rough terrain but are computationally expensive to balance and control.
- Aerial Robots (UAVs): Drones. They operate in 3D space and are dominated by the challenges of flight dynamics and obstacle avoidance.
- Swarm Robots: Multiple small, simple robots that coordinate collectively, inspired by insect colonies. They offer redundancy i.e. if one fails, the swarm continues.

Robot Classification

B. By Application Domain

- Industrial Robots: Used in manufacturing. High precision, high speed & can perform repetitive tasks.
- Service Robots: Assist humans in daily tasks (e.g., vacuum cleaners, delivery bots).
- Medical/Surgical Robots: High dexterity, sub-millimeter precision (e.g., Da Vinci system).
- Exploration Robots: Operate in extreme environments such as deep sea or deep space, where latency prevents real-time human teleoperation.

AI Components in Robotics

- AI gives robots four major capabilities.

A. Perception

- Perception allows robots to understand their surroundings using sensors.
- Common Sensors are Camera , LiDAR , Ultrasonic sensor, Infrared sensor, GPS, Microphone
- AI Techniques used for collecting perception are Computer Vision, Image Processing, Object Detection & Face Recognition.
- Example: A self-driving car detects pedestrians, traffic lights, road signs & other vehicles using perception.

B. Decision Making

- After collecting information, the robot decides what action to perform.
- AI Techniques used are Rule-Based Systems , Expert Systems, Decision Trees & Reinforcement Learning.
- Example: Robot sees obstacle ahead and makes decisions such as Stop, Turn left or Turn right.

AI Components in Robotics

C. Learning

- AI enables robots to improve through experience.
- Types of Learning used are supervised learning, unsupervised learning & reinforcement Learning.
- Example: A warehouse robot initially takes 10 minutes to deliver an item. After repeated deliveries, it learns shorter routes and reduces delivery time to 6 minutes.

D. Motion Planning and Control

- The robot must execute the chosen action.
- Tasks include such as path planning, obstacle avoidance, speed control, manipulation of objects.
- Common Algorithms used are:
 - A* Algorithm
 - Dijkstra's Algorithm

Robotics Applications

1. Industrial Automation

- Used in industries for Assembly , Welding, Packaging, Painting, etc.

2. Surgical Assistance

- Used for Robot-assisted surgery, Microsurgery, Remote surgery
- Surgery done using robots have high precision, small incisions & faster recovery.

3. Space Exploration

- Robots such as Mars Rover, Lunar Rover & Space Station robotic arms are used for space exploration.

4. Service Robots

- Used to assist humans such as hotel robots, cleaning robots & delivery robots.

5. Humanoid Robots

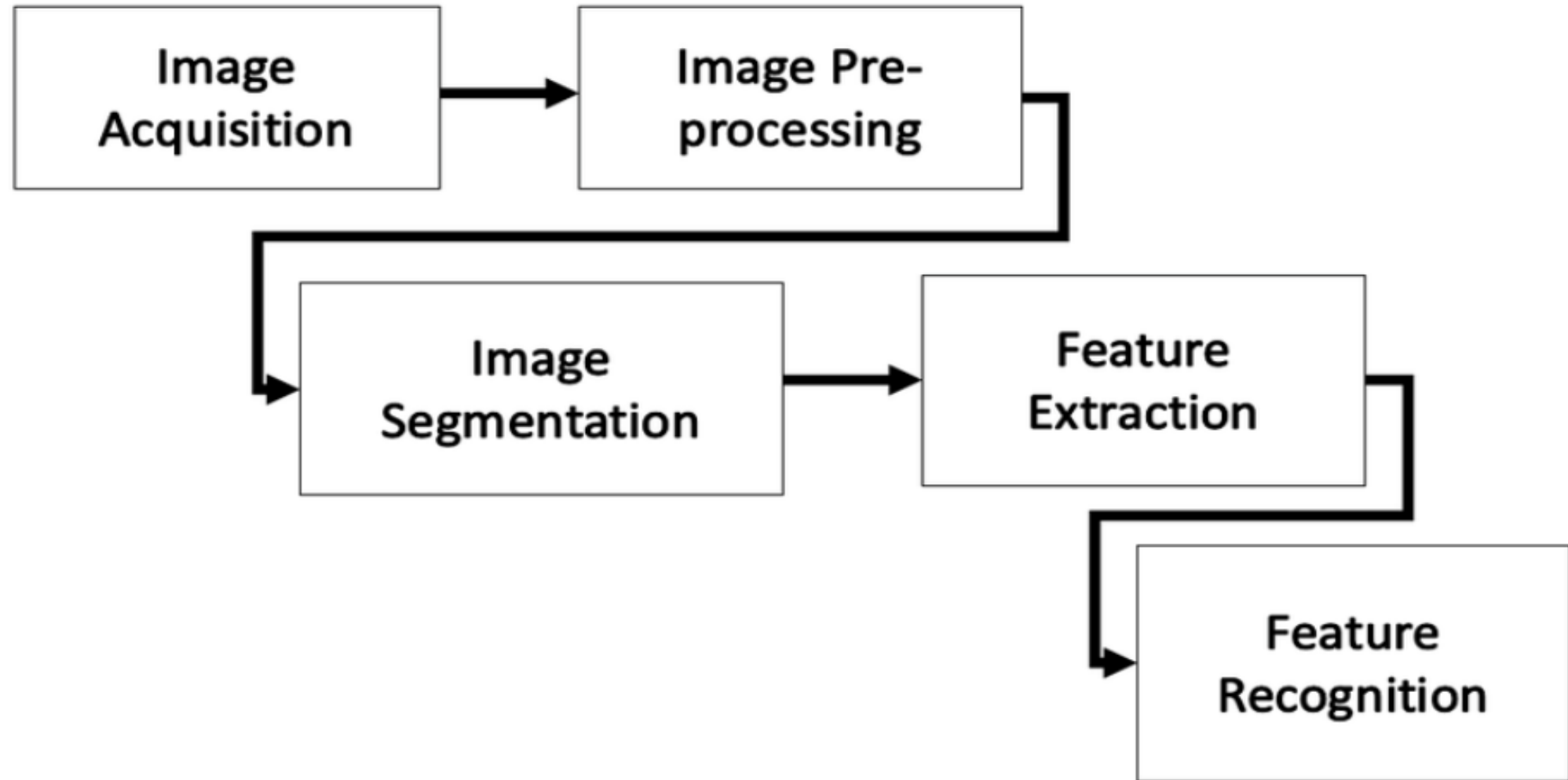
- Designed like humans with capabilities such as walking, speaking & human interaction.

6.3. Robotics & Computer Vision: Fundamental, Components and Applications

- Computer Vision

- Computer Vision is a branch of AI that enables computers to interpret and understand visual information from images and videos.
- Computer Vision takes a 2D image (or video) as input and outputs meaningful information such as object labels, 3D coordinates, or actions.
- The goal of Machine Vision is to create a model of the real world from images.

Core Concepts of Computer Vision/ Computer Vision Pipeline



Computer Vision Pipeline

1. Image Acquisition

- Process of collecting raw data with digital cameras, webcams, CCTV systems, smartphones, drones, satellites, X-ray, MRI or microscopy devices.
- We use still images or video frames considering lighting, object position, environmental factors, sensor calibration, frame rate and image resolution at this stage.

2. Preprocessing

- Raw data is altered to clean up anomalies and normalize input sizes before sending them to a model.
- Operations performed are:
 - Noise Reduction: Applying filters (e.g., Gaussian filter) to remove noise from the image.
 - Normalization: Adjusting the intensity values to a common scale, often between 0 and 1.
 - Image Scaling: Resizing images to a fixed dimension required by the model.
 - Data Augmentation: Techniques like rotation, flipping, cropping and color adjustments to artificially expand the dataset.

Computer Vision Pipeline

3. Image Segmentation

- The image is partitioned into multiple meaningful segments or pixel groups to isolate areas of interest. Some operations performed are:
 - Thresholding: Converting grayscale images into binary (black and white) maps based on pixel intensity.
 - Edge Detection: Running mathematical operators like Canny, Sobel, or Laplacian filters to outline contours.
 - Semantic Segmentation: Utilizing deep networks to label every individual pixel in an image

Computer Vision Pipeline

4. Feature Extraction

- Distinct structural identifiers are collected to help the machine grasp what makes an object unique.
- For modern pipelines, extract multi-scale features with convolutional neural networks, often leveraging transfer learning from pretrained models.
- Keypoint Detection: Identifying key points of interest in the image, such as corners or blobs, using algorithms like SIFT, SURF or ORB.
- Descriptors: Creating feature descriptors that represent the local neighborhood of key points.
- Deep Learning Features: Using convolutional neural networks (CNNs) to automatically learn and extract features from images such as edges, corners, texture, shape & color.
- Feature extraction reduces unnecessary information while preserving important characteristics.

Computer Vision Pipeline

5. Object Detection

- Object detection answers "What objects are present?"
- It identifies object class, object location.
- Example: Detecting car, pedestrian, traffic light
- Popular models used are YOLO(You Only Look Once), SSD(Single Shot MultiBox Detector)

6. Object Recognition and Classification

- Detected objects are assigned to classes using algorithms including SVM, k-NN, Decision Trees or deep CNNs.
- Classification Algorithms: Using traditional machine learning algorithms like SVM, k-NN or deep learning models like CNNs.
- Transfer Learning: Fine-tuning pre-trained models like VGG, ResNet or Inception for specific classification tasks.

Computer Vision Applications

1. Autonomous Vehicles

- To perform tasks such as lane detection, traffic sign recognition, pedestrian detection & obstacle avoidance.

2. Facial Recognition

- Used for smartphone unlock, airport security, attendance system.

3. Medical Imaging

- Used for tumor detection, X-ray analysis, MRI analysis & CT scan interpretation.

4. Quality Control

- Industries use computer vision for crack detection, defect detection, product inspection.
- Example: Detecting defective bottles on a production line.

The Critical Role of Deep Learning in This Ecosystem

- We cannot discuss modern Robotics and Computer Vision without acknowledging the elephant in the room: Deep Learning has disrupted both fields.
- For Perception: CNNs and Transformers have replaced hand-crafted features. They are far more robust to variations in lighting, occlusions, and background clutter.
- For Planning: Reinforcement Learning (RL) combined with vision allows robots to learn policies directly from pixels (End-to-End Learning), eliminating the need for explicit programming of every movement.
- For Data Generation: Generative models (GANs and Diffusion models) are now used in the Sim2Real pipeline generating massive, photorealistic synthetic training data to train vision models, bypassing the need for expensive human-annotated real-world data.

6.4 Sustainable AI Systems

- Artificial Intelligence has transformed nearly every industry, including healthcare, transportation, finance, and education.
- However, while AI solves many problems, it also introduces new environmental challenges.
- Large AI models require massive datasets, high-performance GPUs, large data centers & significant electricity.
- This raises an important question: “Can AI itself be developed and used in a sustainable manner?”
- This idea leads to the concept of Sustainable AI Systems.
- Sustainability means meeting present needs without compromising the ability of future generations to meet their own needs.
- A sustainable system aims to balance economic growth, environmental protection & social well-being.

Sustainable AI

- Sustainable AI refers to the design, development, deployment, and use of AI systems in ways that minimize environmental impact while maximizing social and economic benefits.
- It focuses on two complementary perspectives:
 - AI for Sustainability
 - Using AI to solve environmental and sustainability challenges.
 - Sustainability of AI
 - Reducing the environmental footprint of AI itself.
- These two perspectives together form the foundation of Sustainable AI Systems.

Need of Sustainable AI

- AI models are becoming increasingly larger.
- Models such as Large Language Models (LLMs), Autonomous driving systems, Medical diagnosis systems, Climate prediction models are increasingly complex and large.
- These systems require thousands of GPUs, large cloud data centers, continuous power supply & cooling systems.
- This introduces challenges such as high electricity consumption, increased carbon emissions, expensive computational infrastructure & electronic waste (e-waste).
- Therefore, AI should not only be intelligent but also energy-efficient and environmentally responsible.

AI for Sustainability

- AI can significantly contribute to solving environmental problems.

A. AI for Smart Energy Grids

- A smart grid is an electricity network that uses digital technologies, sensors, and AI to monitor and manage electricity generation, distribution, and consumption efficiently.
- AI analyzes electricity demand, weather forecasts, solar energy production & wind energy availability.
- It then predicts future demand and optimizes power distribution.
- Example: On a hot summer afternoon, AI predicts increased air-conditioner usage. It increases electricity supply before demand peaks. This prevents power shortages and reduces energy waste.
- Benefits
 - Lower energy loss
 - Better integration of renewable energy
 - Reduced electricity costs
 - Improved grid reliability

AI for Sustainability

B. AI for Climate Change Prediction

- Climate change involves long-term changes in temperature, rainfall, sea levels, and weather patterns.
- Traditional climate models process enormous amounts of environmental data.
- AI improves this process by analyzing satellite images, ocean temperatures, rainfall records, greenhouse gas concentrations & weather sensor data.
- AI Applications:
 - Flood prediction
 - Cyclone forecasting
 - Drought prediction
 - Wildfire detection
 - Sea-level prediction
- Example: An AI system detects rising temperatures and dry vegetation from satellite imagery & it predicts a high probability of forest fires and alerts emergency authorities.

AI for Sustainability

C. AI for Resource Efficiency

- Resources such as water, electricity, fuel, and raw materials are limited.
- AI helps use them more efficiently.

1. Agriculture

- AI systems predicts irrigation needs, detects crop diseases, recommends fertilizer usage
- This results in less water usage, higher crop yield & reduced fertilizer waste.

2. Manufacturing

- AI monitors machines to optimize production.
- It helps in reduction of material waste, lowers energy consumption & improves production efficiency

AI for Sustainability

3. Smart Buildings

- AI automatically controls lighting, air conditioning & heating.
- Example: Lights automatically switch off when rooms are empty which reduces electricity consumption.

4. Transportation

- AI optimizes traffic signals, delivery routes & public transportation schedules
- This results in less fuel consumption, reduced traffic congestion & lower carbon emissions.

Environmental Cost of AI

- While AI helps sustainability, training modern AI models can consume enormous amounts of energy.

Why Does AI Consume So Much Energy?

- Training large AI models involves:
 - Billions of parameters
 - Millions or billions of training examples
 - Multiple training iterations (epochs)
 - Thousands of GPU hours
- These computations require powerful data centers operating continuously.
- Energy Consumption Sources
 - GPU computations
 - Data storage
 - Network communication
 - Cooling systems
 - Backup power systems

Environmental Cost of AI

Environmental Impact

- High energy consumption can lead to:
 - Increased electricity demand
 - Higher carbon dioxide (CO₂) emissions
 - Greater dependence on fossil fuels (where renewable energy is unavailable)

Sustainability of AI

- Green AI Practices:
 - Green AI focuses on reducing the environmental impact of AI while maintaining acceptable performance.
1. Efficient Algorithms
 - Develop algorithms that require fewer computations, less memory & shorter training time.
 2. Smaller Models
 - Instead of always using the largest model, choose a model appropriate for the task. Examples: MobileNet, TinyML models, Distilled models
 3. Model Compression
 - Reduce model size using techniques such as pruning (removing unnecessary parameters), quantization (using lower-precision numbers) & knowledge distillation (training a smaller model from a larger one)
 - Benefits of model compression are faster inference, lower energy consumption & easier deployment on mobile and embedded devices.

Sustainability of AI

4. Renewable Energy Data Centers

- Use renewable energy sources such as solar energy, wind energy, hydroelectric power to power AI infrastructure.

5. Efficient Hardware

- Modern AI accelerators provide higher performance with lower power consumption.
- Examples: GPUs, TPUs (Tensor Processing Units) & NPUs (Neural Processing Units)

6. Responsible Model Selection

- Instead of training a new model from scratch we can use pre-trained models, fine-tune existing models, share trained models through open repositories.
- This reduces redundant computation and energy use.

Sustainable AI and the UN Sustainable Development Goals (SDGs)

Sustainable Development Goals	AI Contribution
Clean Energy	Smart energy grids and renewable energy forecasting
Climate Action	Climate modeling, disaster prediction, wildfire detection
Zero Hunger	Precision agriculture and crop monitoring
Clean Water	Water quality monitoring & efficient irrigation
Sustainable Cities	Smart transportation & energy-efficient buildings
Responsible Consumption	Waste reduction & optimized resource management

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